Investigating Ukrainian Refugee Wage Assimilation Dynamics in Slovakia

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June 2026

Abstract

This paper investigates the wage assimilation of Ukrainian refugees in Slovakia and attempts to disentagle genuine wage growth from selective exit bias of lower-paid workers from the observed labor market. We find that refugee wage assimilation outpaces labour migrants but for women is associated with increased initial wage penalties. The observed assimilation trajectories are sectoraly robust and not driven by sector switching or selective attrition on observable characteristics. These findings speak to a long-standing literature predicting that refugees integrate more slowly than economic migrants, a prediction our unusually high-skilled predominantly female cohort appears to defy. To investigate, we exploit a novel monthly individual-level panel of Social Insurance contributors from the Slovak Ministry of Finance covering over three million workers from 2017 to 2025. We estimate a Chiswick-style wage assimilation model with cohort-specific intercepts by gender, then use a Cox proportional hazard model of labor market exit to construct inverse probability weights correcting the assimilation estimates for selective attrition.

1 Introduction

1.1 Background

The Russian invasion of Ukraine in February 2022 triggered one of the largest forced displacement events in recent European history [16]. Slovakia, sharing a border with Ukraine, received over one million Ukrainian nationals following the invasion, with more than 110,000 filing for temporary protection status in 2022 alone [17]. Brell et al. [4], present a baseline expectation: refugees integrate more slowly than economic migrants and converge toward native wages only gradually. Forced migration precludes the positive self-selection that characterizes voluntary labor flows [6], and the urgency of displacement limits refugees' ability to invest in destination-specific human capital before arrival, producing weaker labor market networks and greater initial occupational mismatch [4]. Prior studies of the Ukrainian wave in Central and Eastern Europe are broadly consistent with this framework, documenting rapid labor force entry but at a substantial occupational cost, with refugees concentrating in low-skill roles well below their educational attainment, and language emerging as the primary barrier to appropriate skill matching [19, 9, 8].

Against this backdrop, the wage assimilation trajectories we observe in our novel administrative dataset are surprising. Although female Ukrainian refugees enter the Slovak labor market with larger initial wage penalties than pre-invasion Ukrainian labor migrants, their wages converge toward native wages at a substantially faster rate than those of labor migrants. Faster convergence is also observed for male refugees whose initial penalties do not significantly differ from labor migrants.

The most natural explanation is selection bias. When lower-paid workers exit the labor market at high rates, the observed average wage of those who remain rises mechanically, overstating true earnings growth [13, 1]. While labor migrants are tied to Slovak employer-sponsored work permits, Ukrainian refugees enjoy substantially greater occupational and cross-border labor mobility under the European Union's Temporary Protection Directive. We explore selection bias as a possible driver of these results and find that the observed assimilation trajectories are not inflated by lower-paid workers exiting. There is an unprecedented level of human capital in this European refugee wave, with approximately seventy percent of working-age female refugees holding university degrees, [14] which suggests that genuine wage growth and occupational upgrading is the most likely driver of our estimated wage trajectories. Using broad NACE sectoral categories, we also find that sector-switching is not a driver of refugee wage assimilation, leaving within-sector occupational upgrading as the most plausible explanation.

1.2 Research Question and Motivation

This paper aims to investigate the assimilation of Ukrainian refugees towards native Slovak wage rates, and the extent to which they are driven by selective exit bias. Our main avenue of exploration focuses on the distinction between pre-war labor migrants and refugees along the dimension of gender. This separation is motivated by the stark differences in socio-demographic characteristics and the arrival conditions of these cohorts. Using individual level monthly panel data, we first estimate a Chiswick-style wage assimilation model to document raw assimilation profiles across arrival groups. We then estimate a Cox proportional hazard model to characterize the determinants of labor market exit among Ukrainians and use predicted survival probabilities to construct inverse weights that adjust the assimilation estimates for selective attrition. To our knowledge, this is the first study of the Ukrainian refugee wave to use individual-level administrative wage records, enabling us to speak directly to wage dynamics that prior existing work has been unable to observe with longitudinal precision (Guzi et al., 2024; Kosyakova and Brucker, 2024; Postepska and Voloshyna, 2025; Botelho, 2022).

Forced migration of Ukrainians into Eastern and Central Europe represents an unprecedented refugee labor supply shock for the region. This refugee influx is also unique within a broader context of previous refugee flows that motivated relevant wage-assimilation literature. The unique demographics and lingusitic/cultural proximity to host countries, accompanied by unprecedentedly permissive regulatory frameworks encouraging labor force participation, invite a re-evaluation of established notions of migrant wage assimilation and provide a previously unexplored context for these dynamics.

2 Literature Review

2.1 Wage Assimilation and Its Biases

The concern over estimating whether immigrants' wages assimilate to native wages over time has occupied a central role in labor economics for the last 50 years. Chiswick [5], based on cross-sectional data from the 1950s and 1960s, argued that immigrants faced a substantial penalty when entering the labor market but that they observe faster wage growth rates, enabling them not only to catch up but to surpass native wages in 10 to 15 years time.

Borjas [3] was the first to directly challenge this finding, arguing that cross-sectional estimates confound cohorts that are intrinsically different in unobserved quality. Controlling for cohort effects, he found that wage growth is approximately 20 percent smaller than cross-sectional estimates suggested. Lubotsky [13] identifies a further source of upward bias, observing that selective exit of the labor market of lower-earning immigrants causes standard estimates to overstate true wage assimilation by approximately 30 to 50 percent.

Abramitzky et al. [1], studying immigrant wages at the turn of the twentieth century, corroborate these findings. Working off detailed panel data, they show substantial and long lasting heterogeneity among immigrants. Importantly, they find immigrants follow different wage trends as a function of their origins and skill-level but do not converge to native wages. They also show negatively selected return migrants and newer cohorts being low skilled to drives much of the apparent assimilation observed in aggregated data.

A related strand of the literature examines whether refugees integrate differently from economic migrants. Cortes [6], in a study of differential wage using cross-sectional data finds that refugees face an initial wage penalty of approximately 6 percent and work 14 percent less hours relative to labour migrants, attributing this to the absence of positive self-selection under forced displacement. Cortes finds that both population experience substantial wage growth and that refugees make up for the initial penalty in 10 to 15 years after which their wages exceeds that of immigrants. Brell et al. [4], in a survey article, present a more pessimistic picture, asserting that while refugees face larger initial mismatch penalties on entry, their assimilation rate over the medium term is not necessarily faster than those of economic migrants due to a series of factors such as poorer social networks, conflict related health issues, skill mismatch or administrative complexities.

Taken together, the literature suggests three aspects central to our paper: observed assimilation is overstated when cohort differences are overlooked; selective attrition introduces substantial upward bias in measured wage assimilation; and refugees may face poorer initial integration prospects than economic migrants, though the gap may narrow over time. Our paper aims to uncover the true assimilation rate of Ukrainian refugees net of these biases.

2.2 The Ukrainian Refugee Wave in European Labor Markets

The Ukrainian refugee wave is structurally distinct from prior episodes. The demographic composition is without precedent: over 75 percent of arrivals were women and children, and approximately

70 percent of working-age female refugees held university degrees [14, 17]. The wave also arrived under the first-ever activation of the EU Temporary Protection Directive, granting immediate labor market access and freedom of movement, a feature that neither prior refugee flows nor pre-war Ukrainian labor migrants possessed. However, survey evidence suggests secondary migration from Slovakia remained limited, with only 5 percent of refugees intending to relocate to another country as of 2023 [11], and by late 2024, over three-quarters of displaced persons in Slovakia reported planning to remain more than five years [7].

Studies across Central and Eastern Europe consistently find that, despite high educational attainment, strong prior employment, and roughly 70 percent of refugees reporting work experience before the invasion [2], refugees entered the workforce quickly but at substantial occupational cost. Refugees sorted into low-skill roles well below their qualifications, with language barriers and restricted credential recognition as their primary obstacles [12, 9, 8]. Self-reported survey evidence corroborates the scale of this mismatch: approximately 60 percent of employed Ukrainian refugees across the EU work in sectors unrelated to their previous experience, wages remain roughly 50 percent below those of host-country nationals on average, and 42 percent of tertiary-educated refugees are employed in low-skill jobs [18, 10]. For Slovakia specifically, Veselkova and Habel [19] document an overeducation rate of 46.2 percent among Ukrainian refugees and virtually no upward occupational mobility in the first year. More recent survey evidence points to gradual improvement, with nearly two-thirds of employed refugees in Slovakia reporting that they work in roles corresponding to their qualifications by late 2024. Interestingly, a meaningful gender wage gap persisted, with men earning approximately EUR 380 more per month than women on average [7]. Postepska and Voloshyna [15] find similar patterns of gradual improvement in Czechia, with localized effects concentrated among women. Critically, all existing studies remain constrained to employment outcomes, occupational outcomes, and self-reported wages. This empirical gap is what our dataset and strategy directly address.

3 Data

3.1 Dataset

This paper utilizes a monthly, individual-level panel dataset of social insurance contributors, made available through the cooperation of the Ministry of Finance of the Slovak Republic. The dataset covers all native and foreign working individuals in the Slovak labor market from January 2017 to December 2025, totaling 3,043,811 individuals. Among these, 60,336 unique IDs belong to Ukrainians who joined the dataset after the Russian invasion of Ukraine in February 2022, 45,027 of whom are considered to be taking on full-time jobs. The dataset contains individual-level information on monthly wages, nationality, age, sex, region of residence, and economic sector (NACE). Individual-level education is not contained in the dataset.

3.2 Data Processing

We make several adjustments to construct the analysis sample. First, all wages are deflated using monthly Slovak CPI data and expressed in real euros. Because the data records only wages and registered insured days, which are independent of contract hours, we cannot directly distinguish full-time from part-time employment. We thus use daily wages to normalize incomplete months and classify wages below the Slovak minimum wage as part-time and drop these observations, thereby restricting the analysis to full-time employment spells. We then construct several variables for the analysis: daily real wages at the individual's first observation, an age variable calculated from the difference between birth years and observation date years, and a native wage benchmark (the average daily wage of native Slovak workers) against which refugee and migrant wage convergence is measured.

3.3 Data Descriptives

Figure 1 plots monthly arrivals of Ukrainian workers (both migrants and refugees) into the full-time Slovak labor market. Before the February 2022 invasion, arrivals were low and dominated by men but spike in the immediate aftermath, peaking at 720 men and 3,242 women in April 2022 (the large gender disparity is consistent with Ukraine's wartime restrictions on male emigration). Both male and female arrival numbers decline over the following year, but stabilize above their pre-war baseline, with women consistently outnumbering men throughout the post-invasion period.

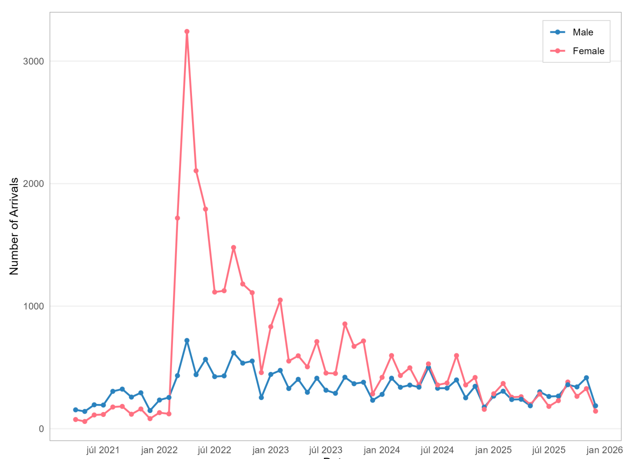


Figure 1: Monthly Arrivals Over Time by Gender

Figure 2 shows the sectors in which Ukrainian refugees first found full-time employment by gender. Both men and women concentrate heavily in administrative and support services and manufacturing. Beyond these sectors, sorting diverges along gender lines, with women clustering in trade (wholesale/retail) and hospitality, while men appear more in transportation and construction. For both genders, entry into high-skill sectors such as ICT (information and communication technology), scientific and financial activities, and education is limited, consistent with occupational downgrading and initial skills mismatch discussed in the literature[1]

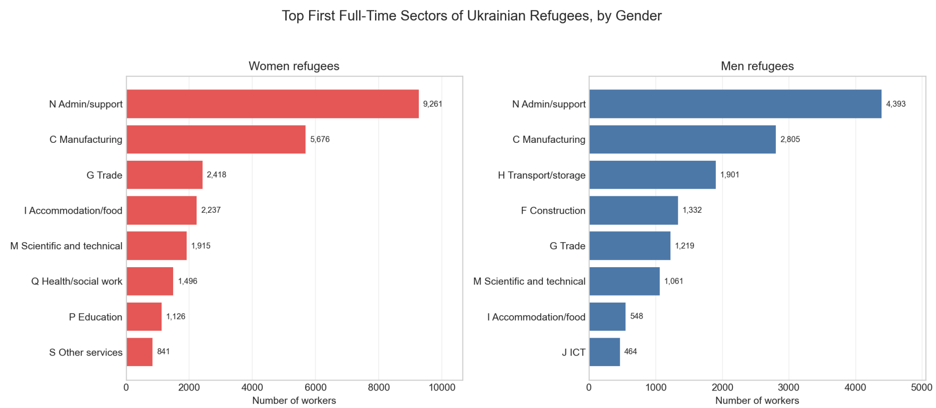


Figure 2: Initial Sector Sorting by Gender

Figure 3 shows the age distribution of Ukrainian refugees at their first full-time job by gender. The Ukrainian government banned emigration for males aged 18 to 60 at the outset of the war, with conscription starting at 25. The spike in males aged 18 to 20 likely reflects young men entering Slovakia to either avoid conscription or to attend university, which exempted them from conscription as long as they followed full-time studies before the age of 23. Women are more evenly distributed, with a modestly pronounced concentration between the late 30s and late 40s.

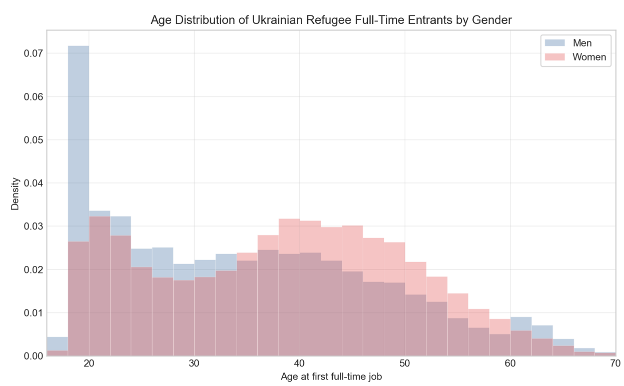


Figure 3: Age Distribution of Ukrainian Refugee Entrants by Gender

Table 1 shows aggregated information on which sectors full-time working Ukrainian refugees sorted themselves in per year. We can see that refugees concentrate in a few sectors such as manufacturing, administrative and support service activities, wholesale and retail or scientific and technical sectors. Comparing Ukrainian and native wages show that refugees systematically earn less than natives even if this gap can substantially vary.

Average Ukrainian wages grow substantially in most sectors and often do so faster than native wages over the same period, suggesting partial convergence. Average wages increased by 29, 19, and 18

[1] See the full classification and sector definitions at `https://ec.europa.eu/competition/mergers/cases/index/nace_all.html` Tables 5 and 6 in the Appendix present a complete breakdown of entry sectors by cohort.

percent in sectors such as, respectively, manufacturing, administrative and support activities, and wholesale/retail. In those same sectors, native average wage growth was only 9, 18, and 10 percent.

It is also important to note that gender composition of the Ukrainian refugees sharply varies across sectors, making close to 90 percent in Education and as low as 30 percent in Construction. Please see the alluvial graph and other tables in the Annex for further descriptives.

Table 1: Ukrainian Refugee Sector Distribution by NACE, 2022–2023 (Average Wages)

2022

NACE	Sector	Ukrainian Refugees	Refugee share of workforce	Ukr. avg wage	Native avg wage	Women (%)
C	Manufacturing	4,335	0.48%	1,035	1,548	76.8%
N	Administrative and support service activities	4,316	2.89%	1,050	1,337	80.0%
M	Professional, scientific and technical activities	1,122	0.64%	1,228	1,926	76.2%
G	Wholesale and retail trade	970	0.19%	1,039	1,396	78.7%
Q	Human health and social work activities	836	0.29%	1,107	1,554	89.4%
I	Accommodation and food service activities	730	1.02%	877	1,011	88.9%
H	Transportation and storage	583	0.23%	1,007	1,368	32.9%
P	Education	499	0.12%	986	1,361	93.6%
J	Information and communication	304	0.24%	1,534	2,422	70.4%
F	Construction	244	0.17%	985	1,405	32.8%
A	Agriculture, forestry and fishing	225	0.25%	1,023	1,312	72.0%
Other	Other	1,031	0.18%	1,192	1,633	86.6%

2023

NACE	Sector	Ukrainian Refugees	Refugee share of workforce	Ukr. avg wage	Native avg wage	Women (%)
C	Manufacturing	4,630	0.87%	1,129	1,556	68.8%
N	Administrative and support service activities	5,189	6.80%	1,108	1,360	73.8%
G	Wholesale and retail trade	1,743	0.53%	1,154	1,422	71.9%
M	Professional, scientific and technical activities	1,511	1.52%	1,321	1,908	70.5%
Q	Human health and social work activities	907	0.53%	1,349	1,630	84.8%
H	Transportation and storage	892	0.55%	1,092	1,396	30.0%
I	Accommodation and food service activities	883	2.38%	948	1,068	82.9%
P	Education	734	0.30%	1,142	1,393	88.6%
J	Information and communication	353	0.43%	2,056	2,246	53.3%
R	Arts, entertainment and recreation	252	0.85%	1,348	1,336	79.8%
F	Construction	325	0.33%	997	1,468	33.8%
A	Agriculture, forestry and fishing	236	0.42%	1,116	1,349	64.4%
Other	Other	990	0.28%	1,178	1,643	80.6%

Notes: Average wage version. NACE S (Other service activities) is pooled into Other. Sectors with fewer than 200 unique Ukrainian refugee IDs in the year are also pooled into Other. Refugee share of workforce is Ukrainian refugee worker-months divided by Ukrainian refugee plus Slovak full-time worker-months in the same sector-year.

Table 2: Ukrainian Refugee Sector Distribution by NACE, 2024–2025 (Average Wages)

2024

NACE	Sector	Ukrainian Refugees	Refugee share of workforce	Ukr. avg wage	Native avg wage	Women (%)
C	Manufacturing	5,232	1.13%	1,235	1,641	65.6%
N	Administrative and support service activities	6,406	8.48%	1,194	1,443	65.2%
G	Wholesale and retail trade	2,071	0.69%	1,188	1,488	71.5%
M	Professional, scientific and technical activities	1,794	1.92%	1,381	2,027	66.8%
Q	Human health and social work activities	1,189	0.71%	1,506	1,736	83.4%
H	Transportation and storage	1,214	0.88%	1,234	1,506	33.4%
I	Accommodation and food service activities	1,069	3.12%	1,021	1,142	80.4%
P	Education	848	0.37%	1,224	1,496	86.8%
J	Information and communication	440	0.57%	2,070	2,335	47.0%
R	Arts, entertainment and recreation	287	1.08%	1,451	1,425	72.1%
A	Agriculture, forestry and fishing	319	0.67%	1,319	1,404	57.7%
F	Construction	379	0.47%	1,106	1,574	36.1%
L	Real estate activities	241	0.74%	1,268	1,715	67.6%
Other	Other	632	0.19%	1,239	1,741	81.5%

2025

NACE	Sector	Ukrainian Refugees	Refugee share of workforce	Ukr. avg wage	Native avg wage	Women (%)
C	Manufacturing	5,981	1.46%	1,332	1,695	63.7%
N	Administrative and support service activities	5,930	7.54%	1,250	1,572	59.4%
G	Wholesale and retail trade	2,211	0.86%	1,223	1,530	71.7%
H	Transportation and storage	1,563	1.32%	1,360	1,556	36.3%
Q	Human health and social work activities	1,362	0.89%	1,686	1,851	82.7%
M	Professional, scientific and technical activities	1,234	1.31%	1,521	2,106	62.6%
I	Accommodation and food service activities	1,121	3.41%	1,096	1,225	77.5%
P	Education	962	0.41%	1,302	1,555	87.6%
J	Information and communication	633	0.90%	2,021	2,548	45.7%
A	Agriculture, forestry and fishing	357	0.90%	1,442	1,450	55.2%
R	Arts, entertainment and recreation	281	1.18%	1,677	1,476	66.2%
F	Construction	297	0.44%	1,245	1,645	32.7%
Other	Other	803	0.24%	1,400	1,805	77.3%

Notes: Average wage version. NACE S (Other service activities) is pooled into Other. Sectors with fewer than 200 unique Ukrainian refugee IDs in the year are also pooled into Other. Refugee share of workforce is Ukrainian refugee worker-months divided by Ukrainian refugee plus Slovak full-time worker-months in the same sector-year.

Table 1: Ukrainian Sector Distribution by NACE on a Yearly Basis

4 Empirical Strategy

The identification in this paper relies on a distinction between Ukrainian arrival cohorts and genders. We employ a gender split sample for all models described in this section to isolate the specific effects associated with the unique structure of these refugee cohorts described in the previous sections. Given that the data is the whole universe of Slovak employment, statistical power is not a limiting factor and this split sampling is preferred to overly complex fully interacted models. Social Insurance data does not allow us to incorporate individual indicators of cohort quality such as education or previous occupation in Ukraine. Refugees are thus split into two waves based on their month of arrival to reflect the expected differences in cohort composition between early arrivals immediately following the Russian invasion and later arrivals. The first wave encompasses arrivals between February 2022 to February 2023 and the second wave comprises arrivals after March 2023, when the influx of new arrivals stabilizes (see Figure 1). Additionally, to allow for comparisons with labor migrants, the third cohort covers pre-war arrivals, excluding left-censored individuals.

The baseline model specification (1) represents a modified Mincerian wage equation with the dependent variable of log daily wages for an individual i in month t. Age acts as a proxy for potential experience with the squared term included to allow for realistic non-linear returns to age. $C_{i,c}$ represents a set of dummy variables for cohorts with natives serving as the omitted reference category. MSA represents months since arrival, taking the value of 0 for the first observed month to allow for the identification of the initial wage penalty. A squared term is also included to allow and test for nonlinearity on the expectation that early wage growth is accelerated by adjustment away from survival jobs, qualification recognition, and language acquisition, and slows down as the individual's labor market position stabilizes.

$$\ln(w_{it}) = \beta_1 \text{Age}_{it} + \beta_2 \text{Age}^2_{it} + \textstyle\sum_c \gamma_{1c} C_{ic} + \sum_c \gamma_{2c}(C_{ic} \times \text{MSA}_{it}) + \sum_c \gamma_{3c}(C_{ic} \times \text{MSA}^2_{it}) + \tau_t + \delta_s + \theta_r + \epsilon_{it} \qquad (1)$$

The baseline specification includes fixed effects for month (τ_t) to absorb seasonal wage fluctuations, region (θ_r) to control for local labor market conditions and sector (δ_s). The inclusion of sector fixed effects is critical to account for the highly concentrated labor market sorting of displaced Ukrainians. Approximately 70% of Ukrainian refugees across all waves and genders initially sort into just four major sector categories. Without absorbing these baseline industry wage differences, the model would conflate the initial penalty with the structural wage discrepancies between these dominant entry sectors and the remaining labor market.

Furthermore, because these sectoral categories are broadly defined, they provide ample room to capture occupational upgrading within the sector so the upward mobility ought to remain correctly captured by the cohort assimilation rate rather than being absorbed by the fixed effect. To explicitly isolate the effect of between-sector mobility, an alternative specification excluding sector fixed effects is estimated.

A hazard model is estimated for each split sex sample of full-time Ukrainian employees. Since we cannot directly observe exits from the labor market or Slovakia itself, we define exits as out-of-observation spells exceeding median length of frictional unemployment, separately averaging for both sexes and the three cohorts. We considered any Ukrainians with observations within that frictional unemployment period from the final date in our data to be right-censored rather than exits.

The hazard models measured the likelihood of exit from the Slovak labor market conditioned on the daily wage of the individual at entry, age and age squared, their region of residence, the sector they worked in at entry, and their entry cohort, interacted with the categorical variables and entry wage. These models were robustness-checked by varying the severity of the right censoring condition and we did not find significant changes in the coefficients, statistical power, or explanatory power of the models.

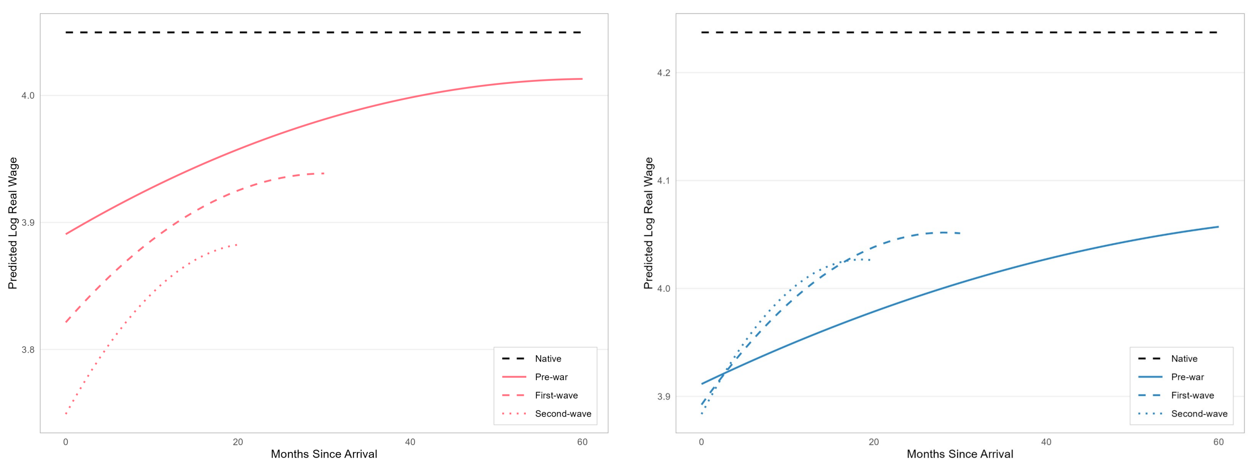
The estimated hazard model is then used to correct for potential selection bias as a driver of observed assimilation using inverse probability weighting. Each migrant observation is weighted by the inverse of its predicted probability of remaining in the sample ($IPW_{it} = 1/\hat{P}(\text{Stay}_{it} = 1)$), while native workers are assigned a constant weight of one. The baseline model is then estimated using weighted least squares (WLS), effectively reweighting the observed sample to counter the potential attrition-driven distortion and isolate the true wage assimilation curve.

For all specifications, standard errors are structurally clustered at the individual level (i). This relaxes the assumption of independent and identically distributed errors, accounting for arbitrary serial correlation and heteroskedasticity within an individual's wage trajectory over time.

5 Results

5.1 Naive Assimilation Model

The baseline specification in Table 2 shows distinct assimilation trends for female and male refugees when compared to their pre-war labor migrant counterparts. Female refugees entering the Slovak labor market face significantly higher initial wage penalties compared to their pre-war counterparts, which is not the case for men. While pre-war female labor migrants entered with an average wage penalty of 13.4%, this penalty increased to 19.9% and 25.2% for first- and second-wave refugee cohorts respectively. However, both male and female refugee wages increase significantly faster than wages of pre-war labor migrants. The estimated assimilation curves for women and men are shown in Figure 4.



(a) Female

(b) Male

Figure 4: Modelled Wage Assimilation by Gender

This gendered distinction in wage penalties accompanied by similar assimilation rates likely reflects the structural composition of observed migration waves. While pre-war male labor migration was predominantly low-skilled and heavily concentrated in manufacturing and transportation and storage sectors, after the war, the restrictions on out-migration for draft-eligible men likely reshaped cohort quality in an opposite direction to that typically observed in the literature, with more high-skill and educated men forming the refugee cohorts.

This is reflected in the sectoral sorting, with sectors such as Administrative and Support Service Activities as well as less migrant-populated sectors like Professional, Scientific and Technical Activities and Information and Communication receiving a much higher share of male refugees compared to labor migrants (Tables 4 and 5 in the Appendix). Significantly larger pre-war cohort wage penalties for men compared to women also suggest that pre-war cohort quality was higher for women than for men. As refugees, both genders were likely taking initial survival jobs and quickly upgrading and assimilating. However, male cohort quality either improved considerably more and is masking the actual increase in wage penalties in our model that omits individual skill covariates, or men faced a looser labor market due to the smaller influx of male refugees, which is likely why male refugees do not appear to suffer from increasing intial wage penalties.

The categorical differences in initial wage penalties and assimilation rates between labor migrants and refugee waves are sectorally robust. For the most frequent sectors that refugees sorted into, assimilation rates of refugees exceed the assimilation rates of pre-war migrants, and the initial wage penalties are also generally larger for female refugees than for labor migrants (Tables 8 and 9 in the Appendix). The estimated differences in assimilation trends are thus not driven by any single sector but rather reflect general dynamics across the labor market.

Table 2: Models 1–4: Wage Penalties and Assimilation by Gender

	With Sector FE		Without Sector FE	
Variable	(1) Female	(2) Male	(3) Female	(4) Male
Age	0.016***	0.036***	0.019***	0.039***
	(0.00019)	(0.00021)	(0.00019)	(0.00023)
Age^2	-0.00019***	-0.00039***	-0.00020***	-0.00044***
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Pre-war Migrant	-0.144***	-0.296***	-0.194***	-0.355***
	(0.005)	(0.003)	(0.005)	(0.004)
First-wave Refugee	-0.222***	-0.317***	-0.295***	-0.403***
	(0.003)	(0.006)	(0.003)	(0.006)
Second-wave Refugee	-0.291***	-0.301***	-0.381***	-0.417***
	(0.003)	(0.004)	(0.003)	(0.004)
Pre-war Migrant × MSA	0.004***	0.003***	0.004***	0.004***
	(0.00029)	(0.00020)	(0.00031)	(0.00022)
First-wave Refugee × MSA	0.008***	0.011***	0.008***	0.011***
	(0.00030)	(0.00062)	(0.00029)	(0.00059)
Second-wave Refugee × MSA	0.012***	0.015***	0.013***	0.016***
	(0.00056)	(0.00080)	(0.00056)	(0.00077)
Pre-war Migrant × MSA^2	-0.00003***	-0.00002***	-0.00004***	-0.00002***
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
First-wave Refugee × MSA^2	-0.00013***	-0.00020***	-0.00013***	-0.00018***
	(0.00001)	(0.00002)	(0.00001)	(0.00002)
Second-wave Refugee × MSA^2	-0.00028***	-0.00041***	-0.00030***	-0.00040***
	(0.00002)	(0.00003)	(0.00002)	(0.00003)
Observations	79,533,821	78,997,594	80,135,926	79,776,869
R^2	0.165	0.195	0.107	0.108
Within R^2	0.005	0.019	0.007	0.021
Month FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	No	No

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Coefficients are reported in log points with the natives serving as the baseline.

Standard errors are clustered at the individual level (in parentheses).

MSA is months since arrival.

The estimates in Models 1 and 2 should be interpreted as lower bounds that do not capture the effects of sector switching, which is absorbed by the sector fixed effects. Unlike labor migrants, who face substantial constraints in changing jobs due to work permit regulations, refugees can move across jobs and sectors much more easily. If refugees initially take more accessible jobs in lower-paying sectors and later transition into sectors that better reflect their qualifications and offer higher wages, this effect will be absorbed by the baseline model, potentially causing it to underestimate the true assimilation rate.

However, the rapid wage assimilation trajectories observed among refugees do not appear to be driven by sector switching. Models 3 and 4 remove the sector fixed effects with virtually no impact on the estimated assimilation rates, which are approximately 0.1 percentage points higher than those obtained in the baseline specifications. It is important to note that this finding does not rule out job switching as a potential driver of rapid assimilation. Upgrading occupations without leaving a sector may still be a major determinant of refugee wage growth, but this process is analogous to promotion within a firm and cannot be identified with the available data.

The increase in estimated wage penalties relative to Models 1 and 2 reflects the fact that Ukrainian migrants sort disproportionately into lower-wage sectors, as documented in the data description section. This sorting is also observed for men.

Results from Models 1 to 4 are also largely robust to the inclusion of part-time workers. While these were excluded from the analysis to avoid imprecise estimations of daily wages, the general trends observed in these models persist even when part-time workers are included. Coefficients for pre-war and first-wave migrants remain robust to this change in sample scope (Table 7 in the Appendix).

5.2 Hazard Model

To test for low-skill selection bias as proposed by **?**], we first estimate hazard models separately for male and female refugees. The specifications condition nonlinearly on age and additionally control for entry wages, region of residence, and sector of entry.

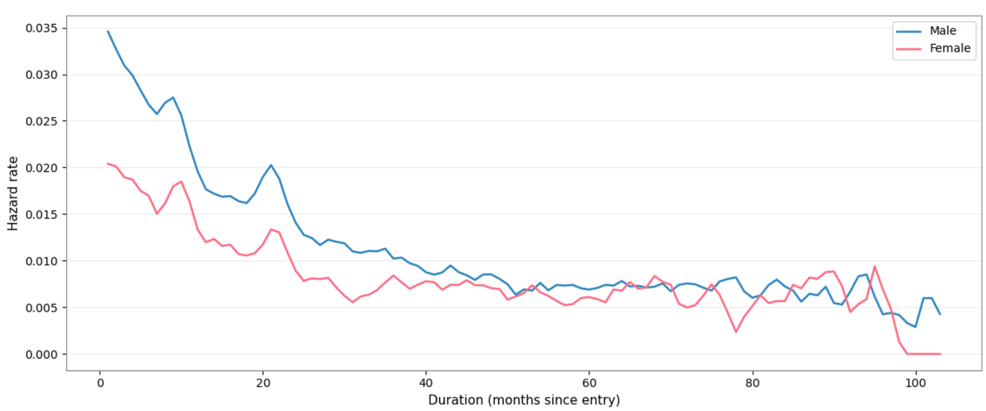


Figure 5: Estimated Hazard Rates for Labor migrants (Pre February 2022)

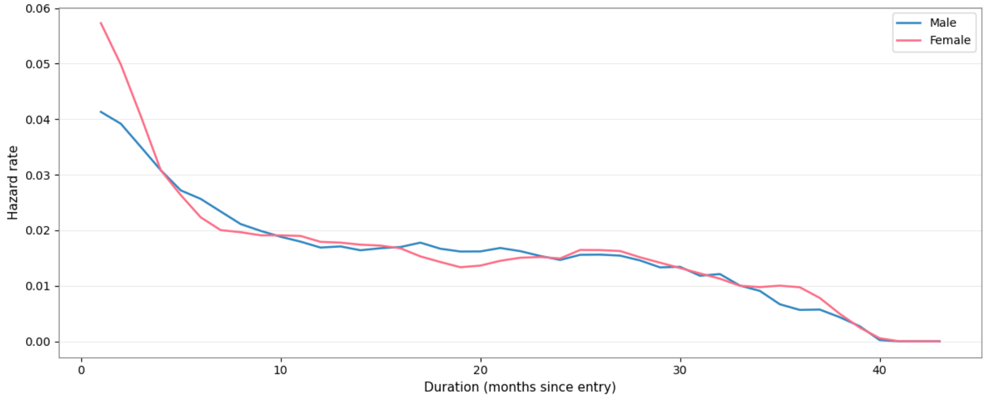


Figure 6: Estimated Hazard Rates for First Wave Refugees (March 2022-February 2023)

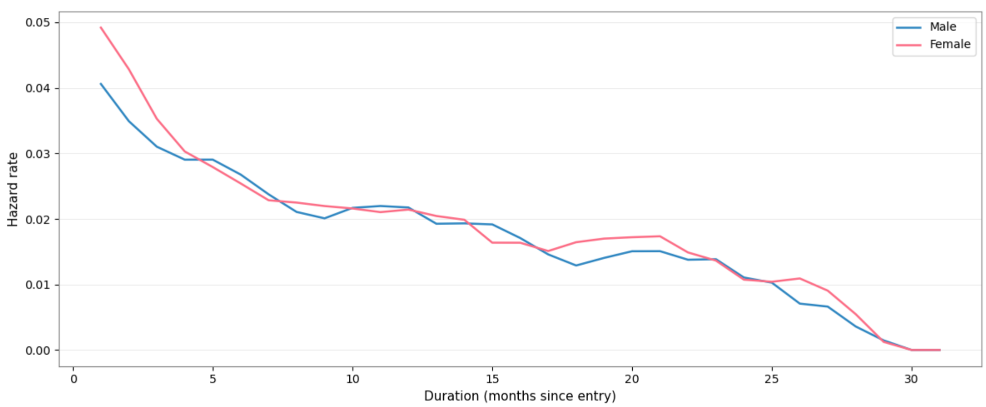


Figure 7: Estimated Hazard Rates for Second Wave Refugees (Post February 2023)

We observe that overall, both male and female refugees leave the Slovak labor market at higher rates than their labor migrant counterparts. As a contrast to labor migrants, where men are less attached to the labor market than women, both refugee men and women display similar hazard rate behavior across the duration of the model.

Table 3: Cox Proportional Hazards Model for Labor Market Exit by Gender

	(5) Female			(6) Male		
Variable	Coef.	Hazard Ratio	p-value	Coef.	Hazard Ratio	p-value
Entry wage	-0.00***	1.00	< 0.005	-0.00***	0.99	< 0.005
Entry wage × Wave 1	-0.10***	1.11	< 0.005	0.02	1.02	0.2
Entry wage × Wave 2	-0.13***	1.14	< 0.005	0.08***	1.08	< 0.005
Age	-0.00	1.00	0.15	-0.01***	0.99	< 0.005
Age^2	0.00**	1.00	0.01	0.00	1.00	< 0.005
Wave 1 cohort	0.13***	1.13	< 0.005	0.00	1.00	0.87
Wave 2 cohort	0.14***	1.15	< 0.005	0.19***	1.21	< 0.005
Observations		37,958			32,325	
Events observed		20,435			18,116	
Partial log-likelihood		-203,809.65			-177,215.28	
Concordance		0.61			0.63	
Partial AIC		407,715.29			354,576.56	

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The specification uses Breslow baseline estimation, a penalizer of 0.50, and no L1 penalty.

Entry-sector indicators, entry-sector interactions with entry cohort, and region-residence indicators, are omitted from the table.,

Our results find that entry wages are not significantly likely to have an effect on labor migrant women, while a single euro increase in the entry daily wages is correlated with a 1% increase in the likelihood of staying for labor migrant men. However, higher entry wages for refugee men and women are actually associated with higher hazards, possibly due to high skill refugees preferring to leave Slovakia for EU countries with stronger labor markets. Because we cannot observe years of education, actual job switching, or actual exits from the country of Slovakia, we cannot claim that this is a causal mechanism, but it is an avenue of further exploration. This unobserved heterogeneity limits the predictive power of our inverse probability weights to only observable drivers in our model, and may obscure other true relationships.

5.3 Selection-Corrected Assimilation Model

The overall estimated wage assimilation effect from our baseline specifications, however, does not appear to be driven by observable selection bias. Models 7 and 8 apply inverse probability weighting to the original specifications from Models 1 and 2, with virtually no effect on the estimated coefficients.

Table 4: Selection-Corrected Assimilation Models 7–8 alongisde Naive Baseline Models 1–2

	Female		Male	
Variable	(7) IPW Corrected	(1) Naive Baseline	(8) IPW Corrected	(2) Naive Baseline
Age	0.016***	0.016***	0.036***	0.036***
	(0.00019)	(0.00019)	(0.00021)	(0.00021)
Age^2	-0.00019***	-0.00019***	-0.00039***	-0.00039***
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Pre-war Migrant	-0.131***	-0.144***	-0.289***	-0.296***
	(0.005)	(0.005)	(0.003)	(0.003)
First-wave Refugee	-0.222***	-0.222***	-0.318***	-0.317***
	(0.003)	(0.003)	(0.006)	(0.006)
Second-wave Refugee	-0.292***	-0.291***	-0.302***	-0.301***
	(0.003)	(0.003)	(0.004)	(0.004)
Pre-war Migrant × MSA	0.003***	0.004***	0.003***	0.003***
	(0.00028)	(0.00029)	(0.00019)	(0.00020)
First-wave Refugee × MSA	0.007***	0.008***	0.010***	0.011***
	(0.00029)	(0.00030)	(0.00061)	(0.00062)
Second-wave Refugee × MSA	0.012***	0.012***	0.014***	0.015***
	(0.00055)	(0.00056)	(0.00079)	(0.00080)
Pre-war Migrant × MSA^2	-0.00002***	-0.00003***	-0.00002***	-0.00002***
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
First-wave Refugee × MSA^2	-0.00013***	-0.00013***	-0.00020***	-0.00020***
	(0.00001)	(0.00001)	(0.00002)	(0.00002)
Second-wave Refugee × MSA^2	-0.00028***	-0.00028***	-0.00042***	-0.00041***
	(0.00002)	(0.00002)	(0.00003)	(0.00003)
Observations	79,521,670	79,533,821	78,983,107	78,997,594
R^2	0.165	0.165	0.195	0.195
Within R^2	0.005	0.005	0.019	0.019
Month FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Ukrainian observations weighted by inverse probabilties from hazard models (5) and (6)
Coefficients are reported in log points with the natives serving as the baseline.
Standard errors are clustered at the individual level (in parentheses).
MSA is months since arrival.

This finding does not imply that there is no heterogeneity in the selection process governing labor market exits among refugees. On the contrary, the hazard models reveal higher exit rates for both male and female refugees associated with each additional euro of daily wages earned at entry, pointing to the possibility of positive selection out of the Slovak labor market. However, by incorporating inverse probability weighting into the assimilation regressions, the estimated wage trajectories are rendered robust to selection on the characteristics we observe. While we cannot rule out selection on unobservable skill because the data does not contain measures of education or pre-war occupation, we can infer that the rapid convergence is not an artifact of the observable selection process, and that the observable seleciton we do detect operates in the direciton that would understate convergence rather than overstate it.

6 Limitations

Our study has several limitations which define avenues for future work. The most consequential concern is our measure of labor market exit. The data set only records formal employment within Slovakia, therefore we cannot observe where a worker goes upon leaving the sample. An exit may reflect return to Ukraine, secondary migration to another EU country, or simply temporary unemployment. We mitigate this by treating only absences longer than the estimated frictional unemployment duration as exits, and our hazard estimates are robust to various severities of this censoring rule. Nonetheless, the interpretation of our results cannot be confirmed as a causal mechanism without observing destinations directly. This remains an avenue for future exploration.

A second limitation is the absence of individual-level human capital measures. The data does not capture information on years of education, specific qualifications, or occupations held in Ukraine prior to displacement, preventing us from directly comparing our results against Borjas [3] cohort-quality framework or from disentangling genuine skill differences across waves from compositional sorting. This prevents us from identifying high and low skill workers through robust measures, especially as wages at entry are not a defensible proxy for quality in this refugee context. We proxy for human capital using sector-level SK-NACE classifications, but lacking occupation classification measures (like ISCO) the firm-level categorization is too broad to accurately capture worker's skill, and cannot distinguish a highly educated worker employed below their qualifications from a genuinely low-skill one.

On a related note, because we observe the sector of the employing firm rather than the specific tasks performed by workers, our estimates cannot identify within-sector occupational upgrading. Such upgrading may represent an important mechanism driving refugee wage assimilation, particularly if workers transition into more complex or better-remunerated occupations while remaining within the same industry. The available data do not allow us to observe these transitions directly, leaving this potentially important channel for future research.

7 Conclusion

This paper studies the wage assimilation of Ukrainian refugees in Slovakia following the Russian invasion of Ukraine. Using a unique monthly administrative panel dataset of Social Insurance contributors obtained through a restricted-access agreement with the Ministry of Finance of the Slovak Republic, we compare the labor market trajectories of refugee cohorts to those of pre-war Ukrainian labor migrants and evaluate the extent to which observed assimilation may be driven by selective exit from the Slovak labor market.

We find that refugee women face larger initial wage penalties (-20% to -25%) than labor migrant women (-12%), while the difference is much smaller for men (-25.9% to -27.4% versus -25.1%). However, refugees of both genders experience substantially faster wage assimilation, with monthly wage growth of 0.7%-1.2% for women and 1%-1.4% for men, compared to only 0.3% for labor migrants.

These findings are not driven by sectoral heterogeneity, heterogeneity between full-time and part-time workers, or sector switching, and are robust to selection out of the Slovak labor market. This suggests that Borjas [3] theory of selection-biased wage assimilation does not explain the fast wage assimilation rates of the Ukrainian refugees.

The most plausible explanation for what these estimates represent is correction for initial skill mismatches. Refugees are likely entering our dataset taking first available survival jobs, incurring high initial wage penalties. As they establish themselves in Slovakia, learn the language, and get their prior credentials recognized, they gradually upgrade into jobs and wages which reflect their true skill.

We also see that refugee women experience higher initial wage penalties compared to refugee men when labor migrants are taken as a baseline; there are two possible explanations for this effect. Firstly, refugee women who in large part emigrated with children may face childcare constraints and face penalties for a job with "child friendly" flexible hours. Secondly, because the refugee wave was so heavily female, Ukrainian refugee women may be competing against other Ukrainian refugee women for the same gender-segregated jobs, driving wages down through labor oversupply.

However, we do not observe specific occupations, qualifications, education levels, or previous work experience in Ukraine, making it impossible to directly measure skill mismatch. Nor can we identify family structures or examine whether childcare responsibilities contribute to wage penalties, which would be avenues for further research.

Our findings suggest that significant deadweight loss is incurred due to initial skill-mismatches in the labor force. Furthermore, there is some evidence that due to EU-wide labor mobility granted to Ukrainian refugees, the highest skilled refugees may also be moving out of Slovakia after an initial period and into stronger labor markets. Policies which accelerate credential recognition, language acquisition, and refugee integration would reduce these inefficiencies and allow refugees to contribute more fully to the Slovak economy.

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8 Appendix

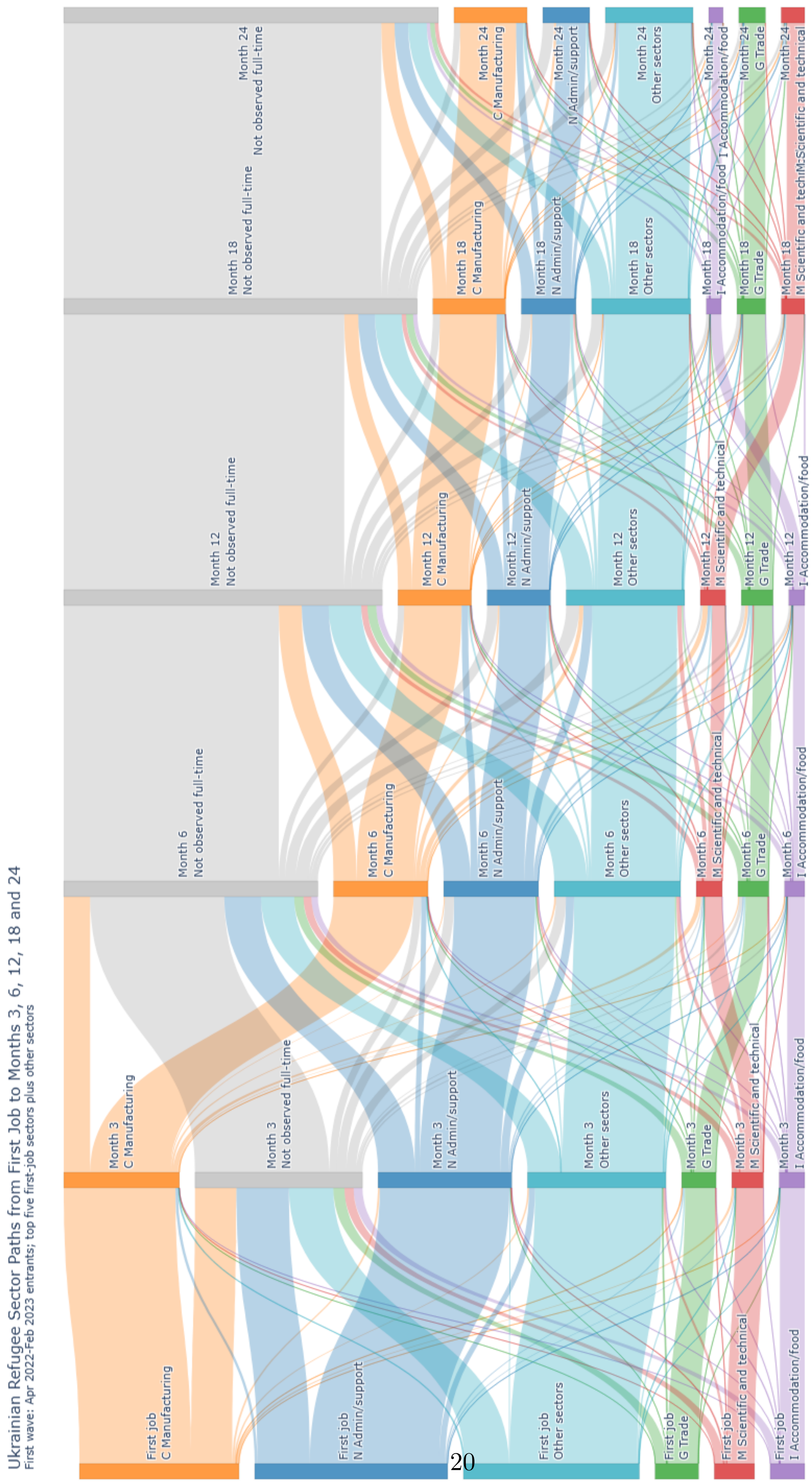

Figure 8: Alluvial Diagram of Sectoral Transitions Among Ukrainian Refugees

Table 5: Female Entry Sector Composition by Arrival Cohort

		Pre-war		First-wave		Second-wave	
NACE	Sector	Count	Share	Count	Share	Count	Share
N	Administrative and support service activities	1,182	16.8%	5,586	32.2%	4,587	35.0%
C	Manufacturing	2,642	37.5%	4,396	25.3%	1,764	13.4%
G	Wholesale and retail trade; repair of motor vehicles and motorcycles	650	9.2%	1,240	7.1%	1,335	10.2%
I	Accomodation and food service activities	494	7.0%	1,140	6.6%	1,252	9.5%
M	Professional, scientific and technical activities	523	7.4%	1,031	5.9%	1,063	8.1%
O	Human health and social work activities	404	5.7%	920	5.3%	631	4.8%
P	Education	140	2.0%	611	3.5%	540	4.1%
S	Other service activities	90	1.3%	567	3.3%	320	2.4%
R	Arts, entertainment and recreation	66	0.9%	408	2.4%	317	2.4%
A	Agriculture, forestry and fishing	198	2.8%	344	2.0%	211	1.6%
H	Transportation and storage	276	3.9%	322	1.9%	358	2.7%
J	Information and communication	123	1.7%	276	1.6%	201	1.5%
O	Public administration and defence; compulsory social security	38	0.5%	164	0.9%	88	0.7%
F	Construction	108	1.5%	156	0.9%	227	1.7%
L	Real estate activities	86	1.2%	136	0.8%	165	1.3%
K	Financial and insurance activities	22	0.3%	27	0.2%	36	0.3%
E	Water supply; sewerage, waste management and remediation activities	2	0.0%	21	0.1%	15	0.1%
D	Electricity, gas, steam and air conditioning supply	2	0.0%	5	0.0%	5	0.0%
B	Mining and quarrying	3	0.0%	3	0.0%	5	0.0%
U	Activities of extraterritorial organisations and bodies	0	0.0%	0	0.0%	1	0.0%
	Total	7,049	100.0%	17,353	100.0%	13,121	100.0%

Shares are calculated within each female Ukrainian arrival cohort.

Table 6: Male Entry Sector Composition by Arrival Cohort

		Pre-war		First-wave		Second-wave	
NACE	Sector	Count	Share	Count	Share	Count	Share
N	Administrative and support service activities	1,909	11.2%	1,636	27.0%	3,340	32.5%
C	Manufacturing	5,527	32.4%	1,418	23.4%	1,617	15.7%
F	Construction	1,657	9.7%	632	10.4%	775	7.5%
H	Transportation and storage	4,580	26.9%	632	10.4%	1,367	13.3%
G	Wholesale and retail trade; repair of motor vehicles and motorcycles	960	5.6%	446	7.4%	840	8.2%
M	Professional, scientific and technical activities	808	4.7%	425	7.0%	732	7.1%
J	Information and communication	165	1.0%	178	2.9%	295	2.9%
I	Accomodation and food service activities	373	2.2%	155	2.6%	414	4.0%
Q	Human health and social work activities	240	1.4%	131	2.2%	195	1.9%
A	Agriculture, forestry and fishing	322	1.9%	123	2.0%	195	1.9%
L	Real estate activities	164	1.0%	57	0.9%	123	1.2%
R	Arts, entertainment and recreation	83	0.5%	53	0.9%	112	1.1%
S	Other service activities	23	0.1%	53	0.9%	63	0.6%
P	Education	58	0.3%	51	0.8%	114	1.1%
O	Public administration and defence; compulsory social security	9	0.1%	27	0.4%	19	0.2%
K	Financial and insurance activities	2	0.0%	12	0.2%	20	0.2%
E	Water supply; sewerage, waste management and remediation activities	29	0.2%	12	0.2%	22	0.2%
B	Mining and quarrying	127	0.7%	10	0.2%	19	0.2%
D	Electricity, gas, steam and air conditioning supply	3	0.0%	2	0.0%	4	0.0%
U	Activities of extraterritorial organisations and bodies	0	0.0%	0	0.0%	1	0.0%
	Total	17,039	100.0%	6,053	100.0%	10,267	100.0%

Shares are calculated within each male Ukrainian arrival cohort.

Tables 5 and 6 summarize the initial sorting of fulltime Ukrainian workers by their arrival cohort and the respective shares of each cohort. An important distiction is in the top 4 sectors with positions 3 and 4 occupied by Wholesale and retail and Accomodation and food sectors for women, whereas for men, Construction and Transportation and storage are much more prevalent.

Table 7: Models 9–12: Wage Penalties and Assimilation by Gender Incl. Part-Time

	With Sector FE		Without Sector FE	
Variable	(9) Female	(10) Male	(11) Female	(12) Male
Age	0.063***	0.071***	0.072***	0.078***
	(0.00026)	(0.00029)	(0.00027)	(0.00030)
Age^2	-0.00076***	-0.00083***	-0.00084***	-0.00091***
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Pre-war Migrant	-0.107***	-0.222***	-0.180***	-0.264***
	(0.007)	(0.004)	(0.008)	(0.005)
First-wave Refugee	-0.299***	-0.401***	-0.461***	-0.549***
	(0.005)	(0.008)	(0.005)	(0.009)
Second-wave Refugee	-0.530***	-0.517***	-0.708***	-0.673***
	(0.007)	(0.008)	(0.007)	(0.008)
Pre-war Migrant × MSA	0.004***	0.003***	0.004***	0.004***
	(0.00041)	(0.00025)	(0.00045)	(0.00027)
First-wave Refugee × MSA	0.008***	0.014***	0.009***	0.016***
	(0.00055)	(0.001)	(0.00053)	(0.00097)
Second-wave Refugee × MSA	0.022***	0.032**	0.024***	0.034***
	(0.001)	(0.001)	(0.001)	(0.001)
Pre-war Migrant × MSA^2	-0.00003***	-0.00002***	-0.00003***	-0.00002***
	(0.00001)	(0.00000)	(0.00001)	(0.00000)
First-wave Refugee × MSA^2	-0.00011***	-0.00021***	-0.00010***	-0.00019***
	(0.00001)	(0.00002)	(0.00001)	(0.00002)
Second-wave Refugee × MSA^2	-0.00052***	-0.00080***	-0.00055***	-0.00077***
	(0.00005)	(0.00005)	(0.00005)	(0.00006)
Observations	91,396,388	90,983,821	92,159,016	91,942,432
R^2	0.195	0.188	0.101	0.089
Within R^2	0.054	0.044	0.055	0.048
Month FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	No	No

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Coefficients are reported in log points with the natives serving as the baseline.

Standard errors are clustered at the individual level (in parentheses).

MSA is months since arrival.

Table 7 replicates Table 2 from the result section with the inclusion of part-time workers (earning above a week of minimum wage in the observed month). The general trend observed for women holds even with this extension and the across the board increase in wage penalties for refugees reflects the limitations of the part-time worker data. Since computed daily wages cannot be adjusted for actual days worked and are calculated by days registered for social insurance which are based on contract dates, daily wages for part-time workers are artificially much lower. A higher prevalence of part time workers among refugees thus inflates the wage penalty against natives.

A distinct feature of this whole universe model is higher assimilation rates for the last wave of refugees which likely reflects increases in labour supply of part-timers driving the a large share of the effect for a cohort with the shortest observed time. Increasing wage penalties also emerge for men as an artifact of the disproportionate presence of part-time workers in refugee cohorts compared to migrants.

Table 8: Female Assimilation Models by Sector

	Administrative and support service activities	Human health and social work activities	Manufacturing	Professional, scientific and technical activities	Wholesale and retail trade; repair of motor vehicles and motorcycles	Other
Age	0.020***	0.011***	0.019***	0.039***	0.018***	0.025***
	(0.00068)	(0.00056)	(0.00044)	(0.001)	(0.00043)	(0.00028)
Age^2	-0.00029***	-0.00013***	-0.00023***	-0.00049***	-0.00022***	-0.00026***
	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00000)
Pre-war Migrant	-0.196***	-0.099***	-0.084***	-0.329***	-0.284***	-0.354***
	(0.010)	(0.023)	(0.006)	(0.018)	(0.015)	(0.010)
First-wave Refugee	-0.183***	-0.323***	-0.227***	-0.256***	-0.321***	-0.350***
	(0.005)	(0.013)	(0.005)	(0.015)	(0.010)	(0.006)
Second-wave Refugee	-0.234***	-0.373***	-0.332***	-0.475***	-0.336***	-0.417***
	(0.005)	(0.016)	(0.008)	(0.012)	(0.009)	(0.006)
Pre-war Migrant × MSA	0.003***	0.006***	0.003***	0.008***	0.007***	0.006***
	(0.00076)	(0.001)	(0.00037)	(0.00096)	(0.00080)	(0.00055)
First-wave Refugee × MSA	0.011***	0.010***	0.007***	0.014***	0.010***	0.007***
	(0.00066)	(0.002)	(0.00055)	(0.002)	(0.001)	(0.00074)
Second-wave Refugee × MSA	0.016***	0.015***	0.012***	0.012***	0.013***	0.009***
	(0.00096)	(0.003)	(0.001)	(0.003)	(0.002)	(0.001)
Pre-war Migrant × MSA^2	-0.00001	-0.00003**	-0.00003***	-0.00007***	-0.00006*	-0.00005***
	(0.00001)	(0.00002)	(0.00001)	(0.00001)	(0.00001)	(0.00001)
First-wave Refugee × MSA^2	-0.00024***	-0.00009**	-0.00009***	-0.00038***	-0.00015*	-0.00014***
	(0.00002)	(0.00004)	(0.00001)	(0.00005)	(0.00003)	(0.00002)
Second-wave Refugee × MSA^2	-0.00048***	-0.00031***	-0.00022***	-0.00027**	-0.00030***	-0.00017***
	(0.00004)	(0.00012)	(0.00005)	(0.00011)	(0.00006)	(0.00005)
Num.Obs.	2,808,091	10,458,532	14,414,259	3,584,761	12,074,199	36,796,084
R2	0.132	0.118	0.143	0.125	0.116	0.112
R2 Within	0.030	0.004	0.007	0.023	0.008	0.012
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Coefficients are reported in log points with the natives serving as the baseline.

Standard errors are clustered at the individual level (in parentheses).

MSA is months since arrival.

Table 9: Male Assimilation Models by Sector

	Administrative and support service activities	Manufacturing	Professional, scientific and technical activities	Transportation and storage	Wholesale and retail trade; repair of motor vehicles and motorcycles	Other
Age	0.024***	0.043***	0.051***	0.017***	0.042***	0.039***
	(0.00066)	(0.00036)	(0.001)	(0.00061)	(0.00062)	(0.00039)
Age^2	-0.00032***	-0.00049***	-0.00060***	-0.00017***	-0.00046***	-0.00043***
	(0.00001)	(0.00000)	(0.00001)	(0.00001)	(0.00001)	(0.00000)
Pre-war Migrant	-0.232***	-0.208***	-0.501***	-0.360***	-0.340***	-0.456***
	(0.009)	(0.005)	(0.019)	(0.005)	(0.017)	(0.011)
First-wave Refugee	-0.186***	-0.368***	-0.381***	-0.403***	-0.273***	-0.521***
	(0.011)	(0.010)	(0.027)	(0.016)	(0.020)	(0.013)
Second-wave Refugee	-0.193***	-0.392***	-0.482***	-0.410***	-0.305***	-0.505***
	(0.006)	(0.010)	(0.018)	(0.010)	(0.014)	(0.010)
Pre-war Migrant × MSA	0.001**	0.003***	0.010***	0.004***	0.00061	0.006***
	(0.00070)	(0.00033)	(0.001)	(0.00028)	(0.00094)	(0.00067)
First-wave Refugee × MSA	0.012***	0.012***	0.015***	0.007***	0.004	0.013***
	(0.002)	(0.001)	(0.004)	(0.002)	(0.002)	(0.002)
Second-wave Refugee × MSA	0.016***	0.015***	0.010**	0.014***	0.010***	0.022***
	(0.001)	(0.002)	(0.004)	(0.002)	(0.003)	(0.002)
Pre-war Migrant × MSA^2	0.00000	-0.00002***	-0.00009***	-0.00003***	-0.00001	-0.00003***
	(0.00001)	(0.00000)	(0.00002)	(0.00000)	(0.00001)	(0.00001)
First-wave Refugee × MSA^2	-0.00025***	-0.00021***	-0.00034***	-0.00007	-0.00007	-0.00019***
	(0.00005)	(0.00003)	(0.00011)	(0.00005)	(0.00006)	(0.00004)
Second-wave Refugee × MSA^2	-0.00045***	-0.00033***	-0.00013	-0.00036***	-0.00029*	-0.00051***
	(0.00007)	(0.00006)	(0.00018)	(0.00007)	(0.00012)	(0.00009)
Num.Obs.	3,674,904	25,550,192	3,511,101	8,068,727	9,679,744	29,292,201
R2	0.108	0.151	0.110	0.101	0.093	0.121
R2 Within	0.023	0.027	0.028	0.019	0.022	0.018
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Coefficients are reported in log points with the natives serving as the baseline.

Standard errors are clustered at the individual level (in parentheses).

MSA is months since arrival.

Tables 8 and 9 report results from sectoral split-sample models on a single-sector version of the baseline specification (1) as preented in (2). The superscript s indicates sector, and the sector fixed effect has been removed.

$$\ln(w_{it}) = \beta_1^s \text{Age}_{it} + \beta_2^s \text{Age}_{it}^2 + \textstyle\sum_c \gamma_{1c}^s C_{ic} + \sum_c \gamma_{2c}^s (C_{ic} \times \text{MSA}_{it}) + \sum_c \gamma_{3c}^s (C_{ic} \times \text{MSA}_{it}^2) + \tau_t^s + \theta_r^s + \epsilon_{it}^s \quad (2)$$

There is not a strong indication that the results observed in our paper are driven by any one of the major sectors refugees settled into. In fact, the wage penalty and assimilation trends mirror the overall results of the general model of increasing wage penalties for female refugees and faster assimilation trajectories for both sexes. The only notable exception is the Professional, Scientific and Technical Activities (SK-NACE: M) sector where a clear monotonic increase in penalties and assimilation rates is not observed. Since this is a high-skill sector this likely reflects the fact that specific dynamics of occupational downgrading are less relevant for this sector.

Lastly in the the Manufacturing sector (SK-NACE: C), refugee wage penalties rise even for men, which could reflect high saturation in a sector that absorbed a disproportionate share of refugees of both sexes or a penalty for a genuine skill mismatch for men sorting into manufacturing as a survival job.